(I) using the Method of undetermined coefficients,

Solve

(1) $(D^2+1)y = 12\cos^2 x$; $y = c_1\cos x + c_2\sin x + 6 - 2\cos 2x$

(2) $(D^2+3D+2)y = x+\cos x$; $y = c_1 e^{-x} + c_2 e^{-2x} - \left(\frac{3}{4}\right) + \frac{x}{2} + \frac{1}{10}(\cos x + 3\sin x)$

(3) $(D^2-2D-3)y = 2e^x - 10\sin x$; $y = c_1 e^{-x} + c_2 e^{3x} - \frac{1}{2}e^x + 2\sin x - \cos x$

(4) $(D^2-4D+4)y = x^3e^{2x} + xe^{2x}$; $(c_1+c_2x)e^{2x} + \left(\frac{1}{20}\right)x^5e^{2x} + \frac{1}{6}x^3e^{2x}$

(5) $(D^2+5D+6)y = 2e^{-3x} + 4e^{-2x}$; $y = c_1e^{-3x} + c_2e^{-2x} - 2xe^{-3x} + 4xe^{-3x}$

(II) using the Method of Variation of parameters

Solve

(1) $(D^2+a^2)y = \cot ax$; $y = c_1\cos ax + c_2\sin ax + \frac{1}{a^2}\log\tan\left(\frac{ax}{2}\right)$

(2) $(D^2-3D+2)y = \frac{e^x}{1+e^x}$; $y = c_1e^x + c_2e^{2x} - xe^{2x} + e^x\log(e^{-x}+1) + e^{2x}\log(1+e^{-x}) + \frac{1}{2}e^{2x}$

(3) $(x^2D^2 - xD)y = x^3e^x$; $y = c_1 + c_2x^2 + (x-1)e^x$

(4) $(x^2D^2 + xD - 1)y = x^2\log x,\ x>0$; $y = c_1x^{-1} + c_2x^{-1}\log x + \frac{x^3}{3}\log x - \frac{4x^2}{9}$

(5) $(D^2-2D+1)y = xe^x\log x$; $y = c_1e^x + c_2xe^x + \frac{1}{6}x^3e^x\log x - \frac{5}{36}x^3e^x$

(III) using Reduction of order Method,

Solve

(1) $(x^2+1)y'' - 2xy' + 2y = 0$; $y = c_1x + c_2(x^2-1)$

(2) $(x-1)y'' - xy' + y = 0$; $y = c_1x + c_2e^x$

(3) $y'' - \left(\frac{2}{x^2} + \frac{1}{x}\right)xy' + \left(\frac{2}{x^2} + \frac{1}{2}\right)y = 0$; $y = c_1x + c_2xe^x$

(4) If $y = e^x$ is L.I solution of $xy'' - y' + (1-x)y = 0$ then the other solution is,

a) $-\frac{x}{2}e^{-x} - \frac{e^x}{2}$ b) $\frac{xe^x}{2} - \frac{e^{-x}}{2}$ c) $-\frac{x}{2}e^x + \frac{e^{-x}}{2}$ d) $-\frac{x}{2}e^{-x} - \frac{e^{-x}}{4}$

(5) $(x-1)y'' - xy' + y = 0$; $y = c_1x + c_2e^x$